



Land Type Classification Project Proposal

Project Title

- Land Type Classification - Using Sentinel-2 Satellite Images

Team Members:

1. Abdalaziz Atef Mohammed [Leader]
2. Fatma Osama Abdalaal
3. Abdallah Ashraf Mohammed
4. Maram Mahmoud Mohammed
5. Abdalrahman Mamdouh Rabia
6. Sarah Hagag Mohammed

Internship Track:

- DEPI - Microsoft Machine Learning Engineer

1. Executive Summary

This project focuses on developing a machine learning system for automated land type classification using satellite or aerial imagery. The system aims to classify land into predefined categories such as urban areas, agricultural land, water bodies, forests, and barren land. The solution leverages computer vision and deep learning techniques and follows the full machine learning lifecycle, from data preprocessing to deployment and evaluation.

2. Project Objectives

- Build a reliable land type classification model.
- Apply supervised learning using labeled land imagery.
- Evaluate model performance using clear metrics.
- Deploy a reproducible and scalable ML pipeline.
- Ensure equal contribution across all team members.

3. Problem Definition & Research Question

Research Question:

Can a machine learning model accurately classify land types from satellite imagery within the given project timeline?

Significance:

Land classification plays a critical role in urban planning, agriculture, environmental monitoring, and disaster management. Automating this process reduces manual effort and improves decision-making accuracy.

4. Project Scope

In Scope:

- Dataset selection and cleaning
- Image preprocessing and augmentation
- Model training and evaluation
- Performance analysis
- Documentation and presentation

Out of Scope:

- Real-time satellite feeds
- Commercial deployment
- Multilingual interfaces

5. Methodology, Algorithms & Techniques

- Data Cleaning and Preprocessing
- Exploratory Data Analysis (EDA)
- Feature extraction using CNNs
- Model training using deep learning architectures (e.g., CNNs)
- Hyperparameter tuning
- Model evaluation using accuracy, precision, recall, and F1-score

6. Expected Results / Outcomes

Hypothesis:

The trained model will achieve high classification accuracy across multiple land types.

Expected Outcomes:

- Accurate classification results
- Clear performance metrics
- Confusion matrix and evaluation reports
- Well-documented ML pipeline

7. Project Timeline & Milestones

Phase 1: Project Planning & Management

Phase 2: Literature Review & Requirements

Phase 3: System Analysis & Design

Phase 4: Implementation

Phase 5: Testing & Final Delivery

8. Task Division & Roles

Team Size: 6 Members

Tasks are divided equally across members:

- Data Collection & Cleaning [2: Abdalaziz + Fatma]
- Model Development [3: Maram + Abdallah + Abdalrahman]
- Evaluation & Testing [3: Sarah + Abdalaziz + Fatma]
- Deployment & Visualisation [3: Maram + Abdallah + Abdalrahman]
- Documentation & Presentation [X: Each sub-team presents their phase]

Roles rotate to ensure balanced contribution.

9. Risk Assessment & Mitigation

Risks:

- Poor data quality
- Model underperformance
- Time constraints

Mitigation:

- Early data inspection
- Baseline models first
- Weekly progress reviews

10. Key Performance Indicators (KPIs)

- Model Accuracy (Target $\geq 80\%$)
- Precision & Recall
- Training and inference time
- Milestone completion rate

11. Content Organization & Documentation

All project phases are documented clearly and professionally. The final report includes methodology, analysis, results, and conclusions.

12. Q&A Preparation

The team is prepared to explain:

- Dataset choices
- Model selection rationale
- Evaluation metrics
- Limitations and future work

13. Detailed Timeline Aligned with Internship Deadlines

The project timeline is strictly aligned with the official internship milestones to ensure timely delivery and proper evaluation at each stage.

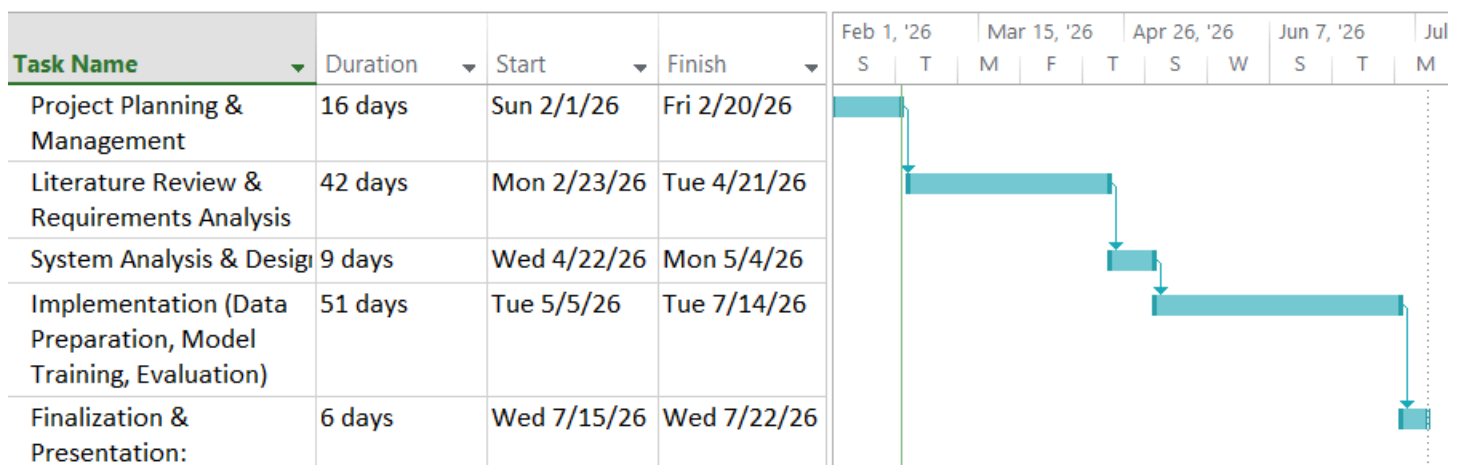
- Project Planning & Management: Now – 20 February 2026
 - Finalize project proposal
 - Assign roles and responsibilities
 - Risk assessment and KPIs definition
- Literature Review & Requirements Analysis: 21 February – 20 April 2026
 - Review related work in land classification
 - Dataset selection and understanding
 - Define functional and non-functional requirements
- System Analysis & Design: 21 April – 1 May 2026

- System architecture design
 - Data pipeline design
 - Model selection strategy
- Implementation Phase: 2 May – 10 July 2026
 - Data cleaning and preprocessing
 - Model training and tuning
 - Evaluation and validation
- Finalization & Presentation: 11 July – 17 July 2026
 - Final report writing
 - Presentation preparation
 - Final system review

14. Gantt Chart Overview

A Gantt chart is used to visualize the project schedule, task dependencies, and parallel work across team members. This ensures balanced workload distribution and progress tracking.

Gantt Chart:



Each phase is assigned clear start and end dates, with overlapping tasks where parallel execution is possible.

Summary

This document outlines a project proposal for land type classification using machine learning, detailing the project's objectives, methodology, timeline, and expected outcomes. It provides a structured approach for an internship or academic submission.